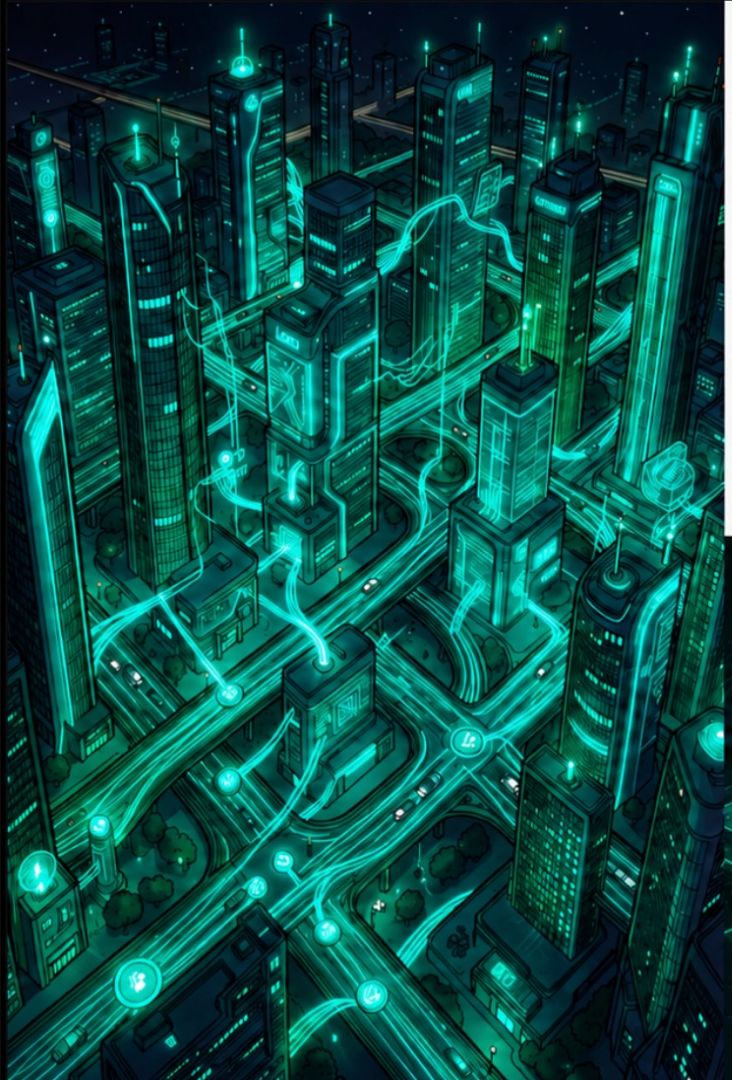


UrbanFlow

Smart Parking Management System



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RFID Access

Real-time Monitoring

Automated Gate Control

SUSTAINABLE.
EFFICIENT. CONNECTED.

Problem Statement



No Real-Time Parking Visibility



Time Wasted Searching for Parking



Traffic Congestion



Fuel & Energy Wastage



No Advance Reservation System



Underutilized Parking Infrastructure

THE PROBLEM

Parking Remains a Major Challenge

⏱ Time Lost

In dense urban areas, drivers spend an average of **17 minutes** searching for parking.

⛽ Fuel Wasted

Up to **30% of urban traffic** comes from vehicles circling in search of available spaces.

🚙 Increased Congestion

Poor parking management contributes to bottlenecks, accidents, and commuter frustration.

🔒 Limited Security

Manual systems lack authentication, audit trails, and real-time visibility.



OUR SOLUTION

Introducing UrbanFlow

A fully automated, IoT-enabled smart parking system that eliminates guesswork – from entry to exit.

RFID Authentication

Instant vehicle identification – no tickets, no queues.

Real-Time Slot Monitoring

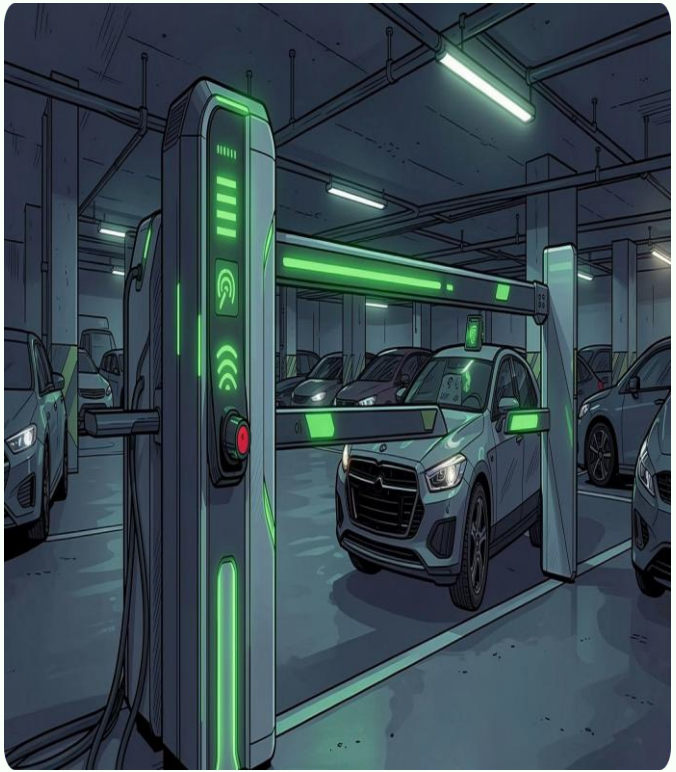
IR sensors track every slot, updating the dashboard live.

Automated Gate Control

Servo motor opens and closes gates on authorized access only.

Smart Dashboard

Live slot count, alerts, and analytics in one clean interface.



UrbanFlow

SOFTWARE WORKFLOW

SENSORS

RFID

IR

DATA TRANSMISSION

PROCESSING & STORAGE

DASHBOARD MONITORING

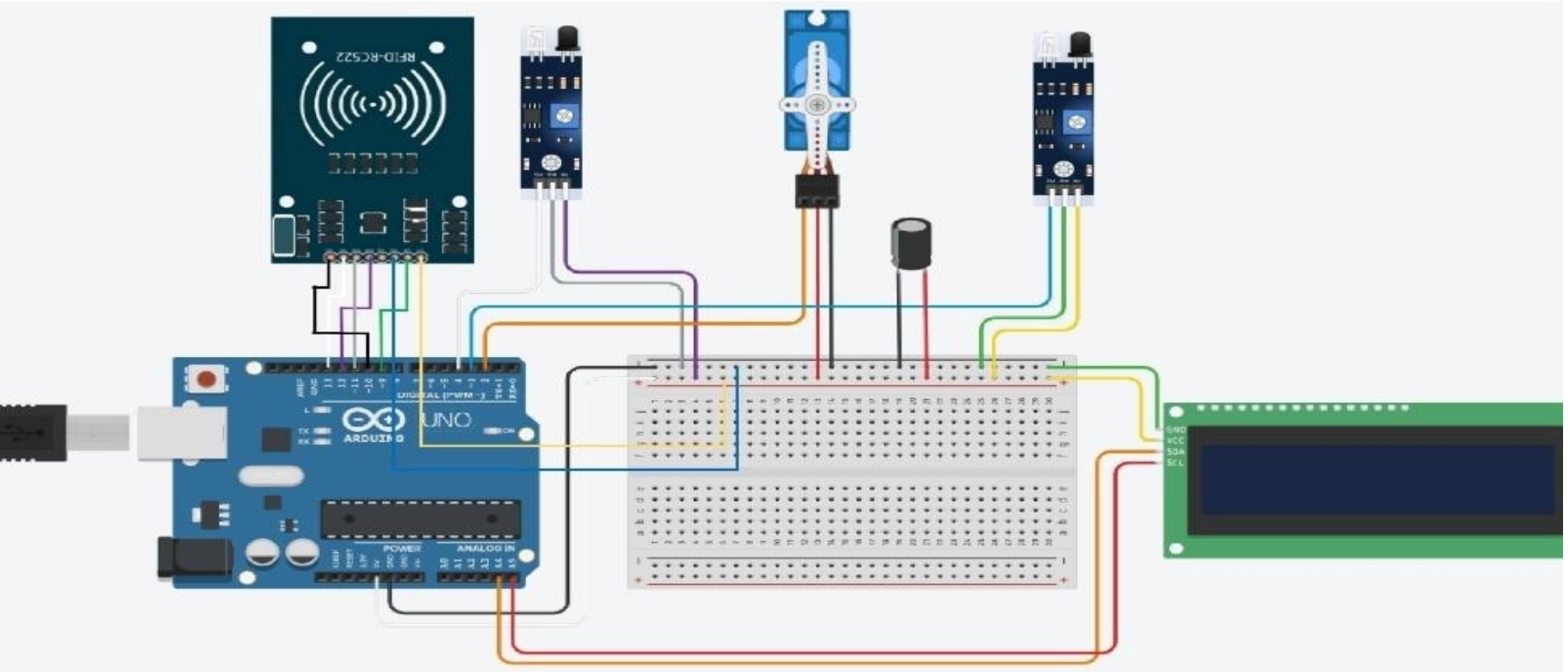
AUTOMATED CONTROL







INTELLIGENT IOT HARDWARE ARCHITECTURE

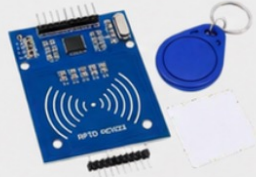


Prototype





Arduino Uno



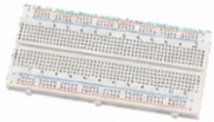
RFID Module



IR Sensor



Servo Motor



Breadboard



16*2 LCD Display



Jumper Wire

LIVE DEMO

Prototype in Action



RFID UID Detection

Instant recognition of registered vehicles — unauthorized tags are denied entry.



Live Slot Count

Dashboard updates in real time as vehicles enter and exit the facility.



Parking Full Alert

System auto-blocks entry and displays "FULL" on LCD when capacity is reached.



Automated Gate

Servo motor responds instantly — no human intervention required.

THE PROBLEM

The Latency Tax Is Killing Urban Mobility-

Today's entry-point management is a **cascade of failure modes**: manual overhead, fragmented data silos, and cloud dependency. Every gate, every checkpoint, every toll booth bleeds time.

⏱ 45s+ Delay

Average manual processing time per vehicle at entry points

☁ Cloud Dependency

Single point of failure — ISP outage means total system blackout

📦 Data Silos

No real-time visibility for city planners or emergency responders



HARDWARE ARCHITECTURE

The Physical Stack



ATmega328P — Logic Core

16 MHz AVR running interrupt-driven firmware. Sub-5ms decision latency on-chip.

MFRC522 — Secure RFID

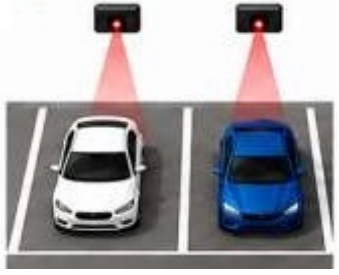
13.56 MHz contactless authentication. Encrypted tag verification at the edge.

IR Sensor Array

Multi-beam obstacle detection. Triggers hardware interrupts on vehicle presence.

System WorkFlow

0



IR sensors detect vehicle presence.

e



Arduino Uno reads sen.sor values.

E)



Atduino sends occupancy data to the l□ptop via USB$erlal c;ommunication.

0



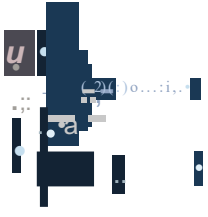
Dashboard software on the laptop updates parking slot status in real time.



When an RFIO card is scanned, Ardu1no v:tlidotes access and controls the servo motor *for* gate operation.



RFID co,d Scanned



Arduino Validates Acceu



Servo Motor Activated



Gate Open$/ Clo$eS



Vehicle Entry/ Exit Successful

REAL-TIME VISUALIZATION

The City Planner's Dashboard



Every gate event streams to a **live WebSocket-fed dashboard** – giving city planners instant situational awareness without polling, without lag.

Live Gate Status

Open/closed state across all nodes, updated in real-time

Vehicle Throughput Analytics

Peak-hour patterns, bottleneck identification, trend lines

Alert & Anomaly Feed

Unauthorized access, sensor faults, and edge-case flags

A Frictionless City Is Not a Dream — It's an Architecture Decision

Urban Flow proves that **autonomous, resilient, low-latency mobility** doesn't require massive cloud infrastructure. It requires the right edge intelligence, the right hardware, and the conviction to put **local logic first.**

WHY URBAN FLOW ?

- LOW COST DEPLOYMENT
- REAL-TIME PARKING VISIBILTY
- REDUCED TRAFFIC CONGESTION
- SCALABLE CITY-WIDE ARCHITECHTURE



WHAT'S NEXT

Future Scope & Impact

Expanding from a smart parking system into a connected urban platform that improves mobility, efficiency, and experience at scale.

1

Mobile App

Enable slot reservation and live navigation.

2

Online Payments

Streamline exit with integrated checkout.

3

AI Analytics

Use predictive insights for occupancy and demand forecasting.

4

5V Power Supply

Support larger deployments with a dedicated power supply unit.

🌍 Deployment Impact

Smart Cities

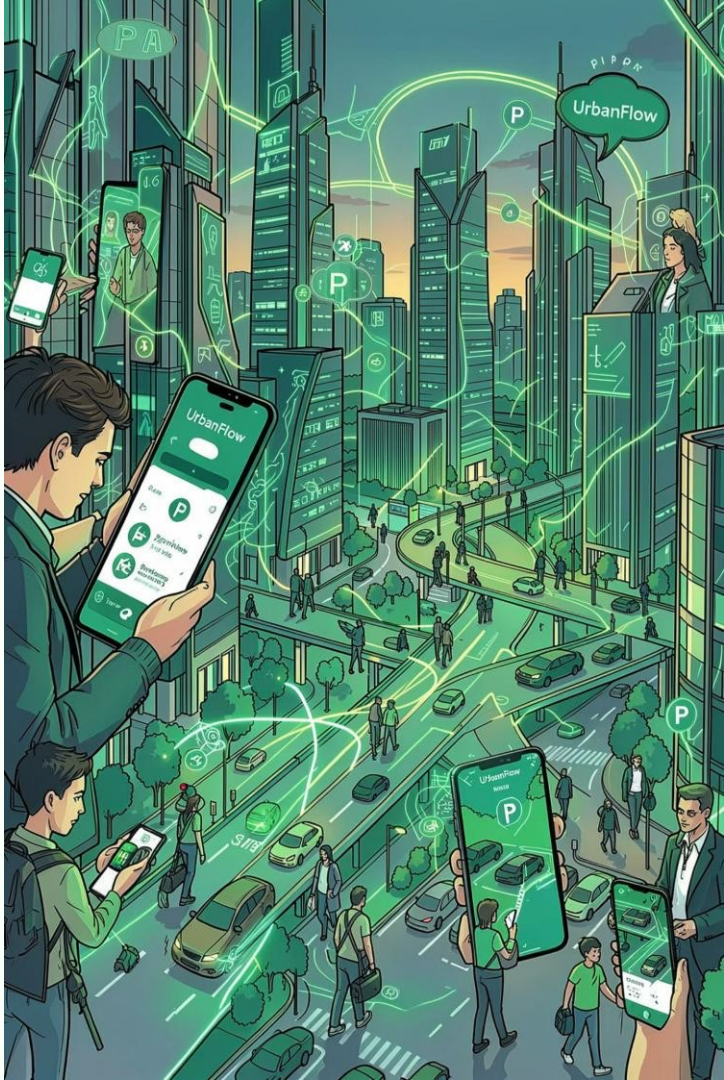
Power city-wide parking intelligence networks.

Hospitals & Malls

Improve mobility and management across high-traffic facilities.

Colleges & Campuses

Deliver secure, student-friendly access experiences.





THANK YOU